Journal of Operations Management

WILEY

RESEARCH ARTICLE

Incentivizing Blockchain Participation Through Task Assignment Mechanisms: Evidence From a Natural Experiment of Consensus Protocols on Ethereum

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Received: 29 November 2023 | **Revised:** 16 February 2025 | **Accepted:** 21 February 2025

Handling Editors: Fabrice Lumineau, Guangzhi Shang, Jay Swaminathan, Gerry Tsoukalas, Stephan Wagner, and J. Leon Zhao

Keywords: blockchain | cryptocurrency | task assignment

ABSTRACT

This study examines how task assignment mechanisms affect the participation of workers on decentralized blockchains. In developing the theory, I highlight that blockchain represents a distinct organizational form for coordinating operations under a highly decentralized structure, in which the essential tasks of system infrastructure maintenance are assigned to third-party crowd workers through the unique governance mechanism of consensus protocol. I specifically focus on two widely adopted consensus protocols in the context of cryptocurrency, namely, proof-of-work (PoW), which assigns tasks that sustain the blockchain system operation based on workers' investments in computing power, and proof-of-stake (PoS), which assigns these tasks based on workers' investments in the native cryptocurrency as stakes. I argue that compared with PoW, PoS increases worker participation and task decentralization because the investment requirement of task participation in the form of blockchain native assets reduces workers' transaction costs in task contracting and their tendencies for hyper-competition. My empirical analysis leverages a natural experiment on Ethereum, namely, the "Merge" event on September 15, 2022, in which the blockchain changed the assignment rules by switching the consensus protocol from PoW to PoS. The results under a difference-in-differences research design confirm my arguments.

1 | Introduction

Over the past decade, the development of blockchains has exerted a substantial influence on many industrial contexts, ranging from supply chains to manufacturing and healthcare (Babich and Hilary 2019; Chod et al. 2020; Hastig and Sodhi 2020; Lumineau et al. 2021). Especially in the financial payment sector, Bitcoin and Ethereum, as the initial applications of blockchain in the form of cryptocurrencies, have grown into trillion-dollar systems with the capacity to process billions of transactions.[1] Accordingly, OM research on technology management has increasingly explored how blockchain shapes operations by enabling secure, reliable, and transparent coordination on a large scale (Hastig and Sodhi 2020; Murray et al. 2021).

The literature has noted that blockchain is not only a generalizable digital technology but also a distinct organizational form with a highly decentralized coordination structure (Lumineau et al. 2021; Wang et al. 2022). Blockchain departs from traditional organizations, as decision and ownership rights are distributed among a wide range of third-party workers who participate in the system operation, rather than being controlled by a central authority (Pagnotta 2022; Zhao et al. 2022). The key mechanism governing blockchain workers is reflected

[Correction added on 23rd August, 2025, after first online publication: The list of handling editors has been updated in this version.]

https://doi.org/10.1002/joom.1358

in "consensus protocols," which are self-executed algorithms that specify how participants of the system reach an agreement state so that blockchain tasks of infrastructure maintenance (i.e., transaction validation) can be coordinated and performed (Biais et al. 2019; Wang et al. 2022). While the OM literature on technology management has examined blockchain applications and how such an organizational form achieves coordination among distributed participants (Kumar et al. 2020; Lumineau et al. 2021), it has not fully explored how the design choices of consensus, as the fundamental governance mechanism, shape participants' motivation and dynamics of blockchain operation.

Against this backdrop, this study examines *how consensus design affects worker participation and blockchain decentralization through the specification of task assignment mechanisms*. My focus on consensus from the perspective of task assignment is primarily motivated by the critical role of task assignment in incentivizing workers and determining operational efficiency underscored in the OM literature on task management (Bai et al. 2022; He and Goh 2022; Liu et al. 2018). Yet, unlike that in centralized organizations, the incentive design of blockchain workers is uniquely defined as specifications in the code programs of consensus regarding the rules for automatic task assignment and corresponding rewards (Lumineau et al. 2021). While the behavioral implications of such distinct design can also be connected to the OM literature on technology management that examines the use of IT and algorithms in task management (Amaya and Holweg 2024; Sampson and dos Santos 2023; Spring et al. 2022), current studies have yet to fully explore those possibilities. My focus on tasks and workers is also derived from the unique blockchain features. Specifically, the tasks to be assigned by consensus pertain to the fundamental infrastructure maintenance rather than complementary activities in centralized coordination (Kumar et al. 2020). Hence, workers' participation and task distribution directly affect the extent to which a blockchain system is sufficiently decentralized to achieve operational efficiency and security. Moreover, blockchain workers are inherently fluid and unconstrained by firm boundaries. Given that the automatic reward distribution as task incentives is concurrently attached to the assignment in consensus, workers can exit or compete for tasks and rewards at any time, solely based on the expected costs and returns inferred from such blockchain governance (Lumineau et al. 2021; Malik et al. 2022). Hence, incentive designs of task allocation and reward rules in consensus determine the extent to which a blockchain can attract and retain sufficient labor to sustain the decentralized structure and efficient operation (Zhao et al. 2022).

To investigate the implications of assignment design for worker participation and task decentralization, I juxtapose two major blockchain consensus protocols used in cryptocurrency blockchains, namely proof-of-work (PoW) and proof-of-stake (PoS), which represent two distinct types of task assignment rules for incentivizing workers. While PoW requires workers to commit a certain amount of computing power to win a task of validating a "block," that is, spaces for documenting verified blockchain transactions (a process referred to as "mining"), PoS assigns such tasks based on the number of native cryptocurrencies pledged by a worker to the blockchain system (i.e., "stakes") (Roşu and Saleh 2021). My key conceptualization draws on the theory of the firm that emphasizes how transaction costs associated with the asset specificity required for task contracting influence the efficiency of blockchain as an organizational form alternative to centralized hierarchies or markets for coordinating economic activities (Lumineau et al. 2021; Wang et al. 2022; Williamson 1985), and on the OM task management literature on the incentive design in crowdsourcing (e.g., Liu et al. 2018; Lo et al. 2024; Ta et al. 2021; Zhao et al. 2022), based on which I propose that PoW and PoS are cases of task assignment that involve distinct contracting processes between workers and the distributed digital system of blockchain. I argue that, in the absence of a centralized coordination authority, blockchain fundamentally shifts the transaction cost of organizing system infrastructure for the key economic activities (i.e., cryptocurrency transactions) to third-party gig workers, through the consensus specifications of pecuniary commitment for task contracting (Gans and Halaburda 2024; Kumar et al. 2020; Wang et al. 2022). PoS increases worker participation compared with PoW because the use of computing power under PoW constitutes a highly specified investment that imposes high search costs and the opportunity costs of task contracting, which aggravates the advantages of centralized organization and incumbent workers through early investment and economies of scale (Arnosti and Weinberg 2022; Easley et al. 2019; Williamson 1985). In contrast, PoS requires the staking of standardized and highly transferable native cryptocurrency for task assignments. Such assignment specification incentivizes greater worker participation by reducing contracting costs and increasing perceived fairness (Roşu and Saleh 2021; Saleh 2021), in turn facilitating task decentralization on blockchains. Drawing on OM studies on how workers' experiences shape their reactions to algorithms (Akşin et al. 2021; Bai et al. 2022; Huckman et al. 2009), I also explore how blockchain workers' experiences affect their sensitivity to task assignment designs and how they differentially contribute to the task decentralization under PoS and PoW.

My empirical examination leverages the exogenous shock of the "Merge" Ethereum upgrade event in September 2022 as a natural experiment, during which the blockchain switched from PoW to PoS, thus altering the task assignment mechanism (De Vries 2023). Using a difference-in-differences (DiD) research design with Bitcoin, which adopted the PoW protocol throughout the sample period, as the comparison group, I find that the switch to PoS through the Merge significantly increases blockchain workers' participation and decreases task concentration on Ethereum. The results also reveal that the incentivizing effects of PoS are more pronounced for new blockchain workers, as predicted by my hypotheses.

Related literature and theoretical contributions: This study makes three contributions to the literature. It first contributes to the OM literature on blockchain applications. This line of research has focused on the adoption of blockchain as an emerging technology in operational settings. A key theme in this literature is that the successful application of blockchain allows firms to increase operational efficiency in external coordination, to satisfy stakeholder demands and expectations through signaling, and to detect hidden market forces and trends (Hastig and Sodhi 2020). Current studies have explored these influences in a variety of contexts including supply chain transparency (Babich and Hilary 2020; Chod et al. 2020), product authenticity/counterfeit detection in retail promotion (Shen et al. 2022; Ying et al. 2023), and operation processes in mining and pharmaceutical industries (Hastig and Sodhi 2020). My study adds to this literature by shifting the focus from blockchain adoption for mitigating external product market challenges to internal governance design that can further increase the efficiency of blockchain application and by uncovering the previously under-addressed potential of blockchain to transform how key resources, such as human capital of crowd workers, are acquired, allocated, and retained. It also contributes to the literature by extending the empirical examination beyond the private centralized operations typically discussed by prior studies, with a focus on public blockchains that emphasize decentralization as a strategic operational goal.

This study also contributes to the second theme emerging from the recent literature investigating the dynamics of blockchain as an organizational form. Lumineau et al. (2021) and Wang et al. (2022) explicitly theorized blockchain as an alternative mode of coordination, which represents a key extension of the classic theory of the firm in explaining the choice of organizational forms for coordinating economic activities (e.g., Coase 1937; Williamson 1981, 1985). Relatedly, several studies have explicitly considered the sources of transaction costs key to the choice of organizational form. Shang et al. (2023) considered how transaction costs from the perspectives of senders and receivers shape cryptocurrency network efficiency. Kumar et al. (2020) proposed that the transaction costs of blockchain adoption could vary across consensus mechanisms, a conjecture that is discussed in detail from the side of blockchain workers in this study. In terms of blockchain governance dynamics, Zhao et al. (2022) examined how the allocation of tasks and rewards affects the performance of a decentralized autonomous organization (DAO) through its token price. Cong et al. (2023) studied how layer 2 scaling enables greater decentralization of public blockchains. Those discussions of blockchain as a distinct organizational form give rise to the theoretical foundations of this study and the focus on governance design. The contribution of my study to this line of research is the explicit focus on consensus as a key governance mechanism influencing blockchain worker participation and decentralization. It advances the understanding of blockchain as an organizational form by explicitly conceptualizing the nature of the blockchain system, the nature of tasks, and the nature of the network, with systematic theorization on the transaction cost of blockchain workers in task assignment and how it shapes governance efficiency.

Relatedly, this study contributes to the research on blockchain consensus protocols. Building on the literature that has examined a single consensus protocol through theoretical modeling, this study provides empirical evidence of the relative efficiency of two major consensus protocols of PoW and PoS. My theory and findings counter the conjecture of the "rich getting richer" effect under PoS, while it is widely documented in the case of PoW (Arnosti and Weinberg 2022; Griffin and Shams 2020; Leshno and Strack 2020; Malik et al. 2022). It hence contributes to this literature by addressing the controversy surrounding the two protocols regarding their effectiveness and efficiency in decentralization (Roşu and Saleh 2021; Saleh 2021). My discussion on the two consensus designs at higher levels of abstraction as generalizable governance mechanisms also enhances the understanding of blockchain governance beyond cryptocurrency.

Second, this study also contributes to the OM literature on task assignments and worker incentives, particularly in crowdsourcing. This literature has explored the division of tasks between gig workers and full-time employees and the matching of tasks to gig workers (e.g., Dong and Ibrahim 2020; Kaynar and Siddiq 2023; Liu, Lou, et al. 2023), as well as incentive mechanisms for gig workers and their behavioral implications (e.g., Gallus 2017; Lo et al. 2024; Ta et al. 2021); this last theme of incentive mechanisms is particularly relevant to this study. For example, Lo et al. (2024) discussed the use of commitment for task bidding as a key incentive on volunteer crowdsourcing platforms, the logic of which is similar to the mining and staking of PoW and PoS for blockchain tasks. Yu et al. (2024) explored the use of auctions to incentivize drivers for on-demand carpooling, which resembles the assignment process as miners essentially use computing power to "auction" for the validation task. Apart from pecuniary incentives, Ta et al. (2021) studied incentives for crowdsourcing task participation in retail operations, highlighting that both intrinsic and extrinsic motivations affect crowd workers' behavior. Earlier research on open-source software communities also uncovered the importance of extrinsic motivation, such as monetary rewards, as well as intrinsic motivation in soliciting contributions from the crowd (e.g., Gallus 2017; Shah 2006). Blockchain validators are also gig workers unconstrained by hierarchies and relational contracts, and their participation is crucial to operation and competitive advantages similar to the literature above (Castillo et al. 2022; Lee and Cui 2024; Miao et al. 2023). The incentive design for centralized crowdsourcing discussed in those prior studies is relevant to my key arguments on how incentives shape blockchain worker participation.

As most of the current studies have focused on the interaction between crowd workers and centralized organizations, this study contributes to this literature by examining workers' behavior under a fully decentralized coordination structure, in which the nature of the tasks shifts from complementary/function (e.g., ridesharing orders) to the fundamental tasks of the core system operation. Relatedly, it also extends the scope of the application of incentive mechanisms highlighted in this literature by theorizing in detail the task assignment dynamics under a new governance mechanism (consensus) in a new organizational form (blockchain). Moreover, this study highlights the conceptualization of blockchain native assets as both assignment rules and rewards in PoS. While this possibility was enabled by the distinct operation of cryptocurrency, it provides implications for future research to study how such task assignment rules can be generalized and applied to other gig economy contexts or broader settings that require decentralized coordination due to globalization and specialization (Alblas 2022; Cui et al. 2021).

Third, this study also relates to the broader OM literature on technology management through IT and algorithms, particularly regarding the behavioral implications of the use of algorithms in operations (Amaya and Holweg 2024; Tan and Staats 2020). My conceptualization of consensus as a task assignment mechanism builds directly on the findings regarding how algorithms affect operational performance by altering workers' behavior through their perceptions of fairness, especially in task assignment (Ai et al. 2023; Liu, Lou, et al. 2023; Yu et al. 2024). At the same time, my findings on the incentivizing effects of PoS reveal the potential of the positive behavioral impacts of algorithms and hence contribute to the literature, which has primarily focused on negative effects, such as algorithm aversion (Dietvorst et al. 2018; Liu, Tang, et al. 2023). Moreover, this study's highlight on the implications of algorithms (i.e., consensus in blockchain systems) beyond IT tools, as key governance strategies for organizing value-creating activities, also creates broader implications for the understanding of the role of AI and other automation technologies in operations, and of how they shape the emergence of new operational strategies and influence the dynamics in traditional and new forms of organizations (Sodhi et al. 2022).

2 | Institutional Background

2.1 | Blockchain and Cryptocurrency: Definitions and Fundamental Features

Defined as "cryptography-based decentralized and distributed systems consisting of an ongoing list of digital records that are shared within a peer-to-peer network" (Wang et al. 2022, 4), blockchain presents several distinct features relevant to this study. In terms of the nature of the system, blockchain constitutes a new organizational form of coordination that fundamentally departs from the typical organizational forms in both operations and ownership (Lumineau et al. 2021; Wang et al. 2022). In terms of the nature of the network and the tasks, blockchain coordinates the essential tasks of system operation and infrastructure maintenance that are traditionally undertaken by a centralized authority through a highly decentralized network structure (Babich and Hilary 2020; Chod et al. 2020; Hsieh and Vergne 2023). These features of blockchain in terms of the nature of the system, the nature of the network, and the nature of the tasks set forth the institutional and theoretical background of my discussion of task assignment through consensus.

I specifically explore how consensus designs shape task participation and decentralization, considering these distinctive features in the context of cryptocurrency. Derived from the invention of Bitcoin by Nakamoto in 2008 and as the earliest and most well-developed application with a multi-trillion-dollar market value as of 2025 (Shang et al. 2023), cryptocurrencies are networks that enable the secure peer-to-peer transfer of natively created digital assets (i.e., coins or tokens) through a decentralized ledger system maintained and coordinated by anonymous and profit-driven third-party workers (also referred to as "validators" or "miners"[2]) (Arnosti and Weinberg 2022; Leshno and Strack 2020; Malik et al. 2022). Essentially, a cryptocurrency blockchain system is composed of two parts: the cryptocurrency as the subject of blockchain transactions (Sockin and Xiong 2023) and the underlying system infrastructure that produces the cryptocurrency and ensures the authenticity, transparency, and traceability of its transactions (Bakos and Halaburda 2022). At the infrastructure level, cryptocurrency transactions are maintained in the unit of blocks—digital storage spaces of a given size for record keeping. The key tasks for workers are to verify, document, and ensure agreement among each other on the accuracy and authenticity of transaction records so that they can be acknowledged by the system (Hinzen et al. 2019; Leshno and Strack 2020). The incentive design governing blockchain workers is also specified by the blockchain consensus. In this study, I draw on the prior literature and define blockchain incentives as a set of automatically implemented rules specified in consensus protocols regarding mechanisms of task assignment and corresponding rewards set to extrinsically motivate blockchain workers' engagement (Benhaim et al. 2023; Lumineau et al. 2021). Cryptocurrencies, in turn, are certificates by the system to reward task completion (Babich and Hilary 2020).

2.2 | Decentralized Blockchain-Based Cryptocurrency and Consensus Protocols

Like blockchain in general, the operation of cryptocurrency relies on consensus protocol as the key governance mechanism, which consists of automated computer codes that dictate the documentation of transactions and the maintenance of the corresponding decentralized ledgers (Biais et al. 2019; Lumineau et al. 2021; Pagnotta 2022). Most cryptocurrencies adopt one of the two mainstream consensus protocols, namely, PoW or PoS. These two consensus protocols specify different assignment rules for validation tasks, that is, for determining who would propose updates of system records for transaction activities as the basis for agreement regarding the true state of the occurrence, timing, and sequencing of blockchain transactions (Amoussou-Guenou et al. 2024; Gans and Halaburda 2024).

PoW was originally created by Nakamoto for Bitcoin as the first consensus protocol (Nakamoto 2008; Amoussou-Guenou et al. 2024). In PoW, validation tasks are assigned based on the computational efforts workers devote to solving a cryptographic problem of identifying a hash value, a unique digital identifier, that meets the requirement set by the consensus so that a new block (a digital space) can be proposed to document transactions (Benhaim et al. 2023). As Nakamoto described in the original Bitcoin white paper, "Proof-of-work is essentially one-CPU-one-vote. The majority decision is represented by the longest chain, which has the greatest proof-of-work effort invested in it" (Nakamoto 2008). In this process, often referred to as hash mining, miners who obtain validation tasks are those that devote greater computing power, as evidence of the commitment they pledge to verify the records for the given task blocks.

PoS, as an alternative consensus protocol, was developed to address the energy consumption issues of PoW and to enhance security, and it has been adopted by several major cryptocurrencies, including Ethereum, Solana, and Polygon (Saleh 2021). In PoS, the tasks of validating a new block of transaction records are assigned based on "stakes," that is, the native cryptocurrency a worker pledges to the network (Amoussou-Guenou et al. 2024; Cong and He 2019; Milunovich 2022). When workers pledge equal stakes, the system assigns the validation task through a random draw. In the case of Ethereum, the PoS protocol requires the deposit of 32 native Ether coins (ETH) as stakes for a worker to serve as a validator.

Both consensus protocols incorporate automated rewards as incentives for validation tasks. Native cryptocurrencies, which convert the rights of an asset into a digital token (Babich and Hilary 2020), are automatically distributed as rewards upon task completion when the network reaches an agreement on the true state of the proposed block, in a fixed amount (for PoW) or with additional fees (both PoW and PoS) (Basu et al. 2023; Shang et al. 2023). This automated reward distribution highlights the importance of task assignment for incentivizing workers as the reward is guaranteed upon completion, without the typical concerns of contract enforcement (Benhaim et al. 2023; Cong and He 2019; Wang et al. 2022).

2.3 | Identification Strategy: The "Merge" of Ethereum

The choice of consensus between PoW and PoS may be endogenous, as it could be related to the strategic purpose and idiosyncrasies of a blockchain (Benhaim et al. 2023), which may also affect worker participation and task decentralization and lead to the potential concern of omitted variable bias in the empirical analysis. To mitigate this concern, I leverage a key upgrade event on a major cryptocurrency blockchain, Ethereum, the Merge in September 2022, during which the system switched from PoW to PoS and thus altered the fundamental task assignment design (Allen 2022).

Apart from the consensus and underlying task assignment rules, the Merge was largely unrelated to other worker-related factors, as the event was carefully designed not to affect the operation of Ethereum (Pavloff et al. 2023), and all technical preparation and testing were initially implemented on an independently operated beacon chain to minimize any potential disruption to Ethereum caused by the transition to PoS. As Figure 1 shows, the price of Ether, the Ethereum native cryptocurrency, the most immediate reflection of the blockchain operations (Zhao et al. 2022), remained largely stable throughout the Merge, making it an ideal exogenous event as a natural experiment for causal inference on the effect of PoS as a task assignment mechanism on the participation and decentralization of blockchain workers.

3 | Theoretical Framework and Hypothesis Development

3.1 | Blockchain as a New Form of Organization and Coordination

In theorizing the effect of task assignment through consensus, I draw on the recent literature on blockchain that takes the perspective of the theory of the firm, which emphasizes the role of transaction costs in determining the efficiency of different organizational forms for coordinating economic activities (Wang et al. 2022; Williamson 1981, 1985), and highlight the nature of blockchain systems as not only a promising enabling technology but also a new organizational form for managing operational and economic activities in ways that fundamentally depart from traditional hierarchical or market structures (Hastig and Sodhi 2020; Hsieh and Vergne 2023). As Lumineau et al. (2021) noted, "blockchains offer a way to enforce agreements and achieve cooperation and coordination that is distinct from both traditional contractual and relational governance as well as from other information technology solutions" as they help "to mitigate cooperation failure at its source potential opportunism in human nature (Williamson 1985) … and offer promising opportunities for facilitating coordination" (Lumineau et al. 2021, 500–508). Hsieh et al. (2018) also noted that blockchains are "non-hierarchical organizations that perform and record routine tasks on a peer-to-peer, cryptographically secure, public network" (Hsieh et al. 2018, 2).

Within a blockchain, the nature of the network that coordinates the system operation represents a fully decentralized structure backed by automated governance (Babich and Hilary 2020). Rather than relying on a central authority for decision-making

FIGURE 1 | Price history of Ethereum and Bitcoin. *Source:* coinmarketcap.com.

typical of hierarchical organizations, blockchain runs on programs of consensus protocols that specify the operational and incentive rules, including automatically executed task assignments, contracts of agreement, and the distribution of rewards (Chod and Lyandres 2023). The distinct decentralized structure of blockchain suggests that the tasks to be coordinated through consensus are related to the infrastructure maintenance of system operation (i.e., validation and ledger keeping), which sustains the performance of functional activities (i.e., financial transactions). Hence, the nature of the tasks to be assigned departs from that under centralized coordination designed to match the functional activities of complementors (e.g., drivers on a ridesharing app or furniture assemblers for TaskRabbit) (e.g., Kaynar and Siddiq 2023; Liu, Lou, et al. 2023). The conceptualization of blockchain, based on transaction cost considerations of organizational forms emphasized by the theory of the firm (Lumineau et al. 2021; Williamson 1985), along with the three distinct features of blockchain, that is, the nature of the network (decentralized structure and ownership), the nature of tasks (infrastructure operation), and the nature of the system (distinct organizational form), forms the theoretical basis of my hypotheses regarding the impact of task assignment through consensus on workers.

3.2 | Task Assignment Mechanisms and Blockchain Worker Participation

While resembling the algorithm-based operational process extensively discussed in the OM literature on task management across organizational forms (Amaya and Holweg 2024; Bai et al. 2022; Kadolkar et al. 2024; Sampson and dos Santos 2023), task assignment through consensus on blockchain faces unique challenges. Efficient blockchain operation requires sufficient decentralization to ensure security and reliability, which naturally demands a large body of workers to undertake the key tasks of validation and ledger keeping (Hastig and Sodhi 2020; Shang et al. 2023). On the one hand, transaction validation is a relatively standardized task that removes the complexity of completion and evaluation (i.e., whether a worker fulfills a task). On the other hand, worker participation is inherently fluid, as the algorithm-based task process enables rewards to be automatically distributed on the completion of each task to anonymous workers unconstrained by organizational hierarchies or relational contracts (Leshno and Strack 2020; Wang et al. 2022). The incentive design in consensus for the task assignment hence becomes critical, as it directly influences the efficiency of the network by affecting workers' decisions of entry and participation.

In essence, PoS and PoW represent a case of task assignment that invokes a distinct contracting process between workers and a distributed digital system without a central authority for coordination. The central principle of task assignment through consensus is reflected in the setup of workers' pecuniary commitment to validation tasks, in the form of hash mining (PoW) or stakes (PoS). While resembling the setup of a "lever" in centralized crowdsourcing to ensure task completion of volunteer workers (e.g., Lo et al. 2024), consensus design is especially important for incentivizing the participation of blockchain workers because, under a fully decentralized structure, such setup fundamentally shifts the transaction cost of system infrastructure maintenance from centralized organizations to individual crowd workers (Gans and Halaburda 2024). As PoS and PoW require distinct types of pecuniary investment as the cost to contract validation tasks with the system for native cryptocurrency in return, they can differentially shape the expectations of rewards as the extrinsic motivation of workers, influencing their participation and task decentralization on blockchains (Cong, Li, and Wang 2021; Malik et al. 2022; Ta et al. 2021).

I argue that compared with PoW, PoS increases workers' task participation, because the nature of the pecuniary investment it specifies lowers workers' participation barriers while providing stronger incentives. From the perspective of transaction cost, consensus protocols specify the investment required to match with validation tasks, which constitutes typical ex-ante transaction costs for workers who seek to contract with the system for assignments and rewards (Lumineau et al. 2021; Murray et al. 2021). Meanwhile, the essential production cost for maintaining the infrastructure pertains only to the labor of verification. Although the automated programs of consensus eliminate the ex-post transaction cost of contract enforcement for validation tasks to a large degree (Lumineau et al. 2021), the contracting process in PoW can still be costly for workers, as hash-mining requires substantial computing power to search for the required random numbers (hash values), leading to high search costs (Wang et al. 2022; Williamson 1985). At the same time, workers in PoW also face opportunity costs because of the high uncertainty in the contracting process for assignments (Arnosti and Weinberg 2022; Shang et al. 2023)—workers who devote the largest amount of computing power have a greater chance of being allocated to validation tasks, while others may only invest slightly less but have a much lower chance to contract with the system, rendering the higher risks all corresponding computing power and costs wasted (Nakamoto 2008).

In contrast, PoS decreases the transaction costs for workers in task contracting and hence lowers the barriers to participation. The pecuniary commitment setup of staking under PoS leverages highly transferable investments. It takes the form of native cryptocurrency created by the blockchain system, which essentially represents the divisible and transferable ownership rights of digital assets as standardized certificates (tokens) acknowledged by the system (Babich and Hilary 2020). The staking setup eliminates the search costs for workers in task contracting as native cryptocurrencies are readily available through trading. The high transferability of such contracting requirements within and outside the system also helps mitigate the opportunity costs of task contracting as workers can recover the costs of stakes by converting them into fiat currencies and gaining additional profit from changes in cryptocurrency prices or using them within the blockchain system for other purposes (Bakos and Halaburda 2022; Saleh 2021; Sockin and Xiong 2023). Moreover, the easy access to standardized stakes as task commitment and random assignment under equal staking eliminates the concern that some workers enjoy a higher chance of assignment by possessing rare commitment resources. It then allows PoS to further incentivize workers by transforming the extrinsic motivation of high-expected rewards with low transaction and opportunity costs into the intrinsic motivation to participate and contribute as a result of workers' sense of fairness in assignment (Bai et al. 2022; Liu et al. 2018; Song et al. 2015). Indeed, such extrinsic motivation combined with intrinsic motivation has been found to incentivize crowd workers by the literature (Gallus 2017; Ta et al. 2021; Yaraghi et al. 2024).

In addition, the high uncertainty in task contracting in PoW based on hash-mining is also likely to trigger workers' hyper-competition and overinvestment, and the associated congestion could further deter participation by aggravating expected transaction costs (Arnosti and Weinberg 2022; Roşu and Saleh 2021). With incomplete information on others' commitment costs and with an oversupply of homogenous labor, workers tend to invest more computing power to ensure successful contracting with the system for task assignment, and the intensified arms race among workers would deter participation (Capponi et al. 2023; Hinzen et al. 2019; Malik et al. 2022). The high transaction costs associated with overinvestment in task contracting in PoW have been noted in the literature. Biais et al. (2019) modeled the PoW consensus as a stochastic game and identified "an arms race in which each miner ends up overinvesting" in computing power. Similarly, Cong, He, and Le (2021) found that the overinvestment in computing power to increase hash rates by a miner can harm others' payoffs and lead to hyper-competition. In PoS, the grounds for excessive spending on computing power are no longer valid (Amoussou-Guenou et al. 2024), and hence it encourages worker participation by mitigating concerns of high transaction cost due to overinvestment and hyper-competition. Those considerations lead to the following hypothesis:

H1. *Workers' task participation on a blockchain under PoS consensus is higher than that under PoW consensus.*

3.3 | Task Assignment Mechanisms and Decentralization of Tasks Among Workers

In this section, I explore the effect of assignment rules in consensus on the decentralization of tasks among workers. Blockchain decentralization incorporates several key considerations regarding the extent to which decision-making rights, rewards, and ownership of infrastructure are equally distributed in the system (Capponi et al. 2023; Lumineau et al. 2021). I focus on the decentralization of validation tasks, defined as the extent to which validation tasks are distributed evenly among blockchain workers (Kwon et al. 2019), as it is the foundation for reward and ownership distribution for overall blockchain decentralization that ensures the reliability, transparency, and security of operations (Chod et al. 2020; Zhao et al. 2022). While decentralization generally relates to the system-level distribution of tasks, it also affects individual workers as the shares of tasks assigned to them relative to the average contribute to the extent of decentralization at the macro level (Arnosti and Weinberg 2022; Capponi et al. 2023).

I argue that PoS facilitates the decentralization of blockchain tasks compared with PoW. Similar to H1, the fundamental consideration pertains to the theory of the firm that focuses on the role of transaction costs in organizing and argues that the asset specificity of the investment required in the contracting of workers for task assignment shapes the efficiency of the decentralized organization (Lumineau et al. 2021; Wang et al. 2022; Williamson 1985). In PoW, investment for task contracting is highly specified as the computing power is based on the specific hash-mining process without alternative uses within or outside the system[3] (Li et al. 2023; Malik et al. 2022). The difficulty of reallocating highly specified asset investment gravitates toward centralized integration, instead of decentralized coordination (David and Han 2004; Williamson 1981, 1985). The concentrated task assignment is more likely in PoW because the costs of hash mining are endogenously different, favoring the lead miners with greater economies of scale and lower transaction costs of task contracting (Capponi et al. 2023; Malik et al. 2022). In contrast, task assignment through staking in PoS only requires highly liquid and divisible native cryptocurrencies (Arnosti and Weinberg 2022; Sockin and Xiong 2023). Such low asset specificity favors a decentralized organizational structure as it lowers the transaction cost of contracting with the system through high liquidity of commitment investment (Bakos and Halaburda 2022; Kumar et al. 2020).[4] At the same time, such greater accessibility of native blockchain assets as stakes allows a wide range of new workers to be eligible for system tasks and for stake holdings to be widely dispersed. This reduces the likelihood that a single entity controls the majority of task assignments by obtaining concentrated shares of stakes (Sockin and Xiong 2023). Indeed, while the literature widely has documented the "rich getting richer" effect for PoW (Cong, Li, and Wang 2021; Malik et al. 2022), it has been noted that the concentrating mechanism through computing power investment does not apply to PoS, as "investors have no incentive to accumulate coins and gamble on the PoS protocol but weakly prefer not to trade" (Roşu and Saleh 2021, 661).

Corresponding to H1, greater worker participation naturally leads to a larger denominator and a lower share of tasks assigned to each worker. The random assignment under equal commitment facilitates decentralization through the even distribution of tasks, as the proportion of tasks previously taken by leading incumbents is reassigned to a greater number of blockchain workers. The redistribution of tasks to the longer tail leads to a smaller distance between the share of tasks taken by each worker relative to the population average, thus further increasing the extent of task decentralization.[5]

The higher level of decentralization under PoS is also in line with the well-established notion in the broader task assignment literature that examines how the fairness of assignment facilitates operational efficiency (Bai et al. 2022; Liu et al. 2018; Tan and Staats 2020). For example, Bai et al. (2022) found that higher perceived fairness increased workers' packing efficiency by over 15% in the context of warehouse management. Similar to H1, the role of fairness in facilitating efficiency also applies to blockchain workers, and the higher level of perceived fairness in the assignment rule of PoS would also promote decentralization as a key indication of the operational efficiency of a blockchain (Cong and He 2019). Hence, I hypothesize:

H2. *Task concentration among workers on a blockchain with PoS consensus is lower than that on a blockchain under PoW consensus.*

3.4 | Worker-Level Dynamics: The Effect of Worker Tenure on Participation and Decentralization

In exploring the heterogeneity among blockchain workers' sensitivity to task assignment design, I focus on how workers' experiences through their tenure on blockchain shape their sensitivity to the different assignment rules of consensus protocols through the mechanisms of H1 and H2. The focus on worker tenure as a moderator is grounded in the OM literature on technology management, which has argued that workers' interactions with algorithms in operational processes are influenced by their perceptions based on their experience (Kadolkar et al. 2024; Rabin 2000). As blockchain coordination and task assignment rely on consensus mechanisms as fully automated algorithms, the dynamics among workers in PoW and PoS may differ as workers' tenure experiences within a blockchain system vary.

I propose that the incentivizing effect of PoS on participation mainly stems from new workers who are more sensitive to the nature of task commitment costs. Under PoW, the transaction costs of contracting with the system for new tasks are perceived as marginal for experienced workers with longer tenure, given that they have already invested in highly specified assets in previous contracting processes. Those workers also have an advantage in the overall average cost of computing power investment, not only because the early block generation has fewer difficulties and higher rewards (Leshno and Strack 2020; Malik et al. 2022) but also because of the economies of scale from knowledge accumulation and familiarity with hardware suppliers and purchasing (Capponi et al. 2023). Experienced miners are thus less sensitive to transaction costs and hyper-competition in PoW. Rather, they benefit from the "rich get richer" effect, given the low investment cost and large economies of scale (Li et al. 2023; Roşu and Saleh 2021).

For newer workers with shorter tenure, the perceived fairness, as the intrinsic motivation derived from the expectation of task allocation and rewards as extrinsic motivation, plays a more prominent role in their decision-making (Gallus 2017; Shah 2006; Ta et al. 2021). Those workers tend to be disadvantaged in PoW based on hash-mining, as they need to invest at a higher cost to contract with the system for assignment, which deters their entry and participation (Malik et al. 2022). In contrast, in PoS, the cost of task contracting is based on the number of stakes and the price of the native cryptocurrency, and the chances of successful contracting do not differ between earlier and later entrants but are strictly based on random assignment under equal stake holdings. This is also perceived as fairer to new workers. The difference between the transaction costs and expected rewards for new workers in PoW is thus greater than that in PoS, translating into a more pronounced incentivizing effect of PoS on worker participation. In addition, the higher accessibility of native cryptocurrency for staking in PoS relative to that of the hardware investment in PoW encourages participation by lowering the entry barrier, which is more relevant to new workers (Benhaim et al. 2023). These considerations lead to a more pronounced increase in the participation of new workers in PoS than in PoW. Therefore, I hypothesize as follows:

H3A. *The positive effect of PoS on task participation is stronger for workers with shorter tenure.*

I also argue that workers with longer tenure may also disproportionally contribute to the decrease in task concentration at the system level. As discussed, the centralizing tendency of task assignment in PoW, due to the high asset specificity of the computing power required, benefits longer-tenured workers. The decrease in contracting costs and increase in the marginal return on investment arising from the economies of scale of computing power allows workers with longer tenure who invest earlier in computing power to receive a disproportionate number of tasks and lead to greater task concentration among incumbent workers (Arnosti and Weinberg 2022). Hence, a shift to PoS from PoW mainly affects the task loads of workers with longer tenure. As the staking commitment with low asset specificity does not differentiate between new and incumbent workers, the cost advantage of early investment is removed, while the likelihood and return on stakes under PoS remain the same for all workers at a given point. This leads to tasks being reassigned mainly from longer-tenured workers with an initially higher task concentration in PoW to new workers with shorter tenure when a blockchain system is in PoS.

In contrast, task decentralization at the system level affects new workers, that is, those with shorter tenure, to a lesser extent. Given a relatively stable level of transaction activities, a higher participation rate under PoS indicates that the average likelihood of a task being assigned to each worker is lower, at a level closer to that of the disadvantaged new workers in PoW. A relatively low likelihood of task assignment suggests that new workers play a smaller role in the change of decentralization, as the chances of being assigned to tasks are relatively low for such workers in both systems. Hence, the differences in the net change in task shares between PoS and PoW are less pronounced for new workers than for experienced workers, whose tasks are more likely to be reassigned in PoS as drivers of decentralization.

As informal interviews with executives of leading cryptocurrency exchanges confirmed, the decrease in decentralization is largely due to the entry of new workers: *Validation becomes much easier as any individual with 32 Ether can just become a node [after the Merge] ... the advantages of previous miners have diminished almost to zero, although knowledge about the network could still be relevant, it is fair to assume that anyone with the [required] number of cryptos [32 Ethers] would have adequate knowledge to be qualified [as a validator]. The likelihood of getting the task would not differ between new workers and those who worked for PoW Ethereum. This is precisely what PoS is designed for, to be fairer and more decentralized*. This corroborates the argument that worker tenure shapes the effect of PoS on the decentralization of tasks among workers. Hence, PoS more drastically affects the task shares of experienced workers with longer tenure, who benefit from higher task concentration in PoW, while the changes in shares of workers with shorter tenure are less pronounced. Hence, I hypothesize:

H3B. *The negative effect of PoS on task centralization is stronger for workers with longer tenure.*

4 | Data and Method

4.1 | Sample Construction and Data Source

As mentioned in the institutional background in Section 2, this study uses a DiD design to derive a stronger causal inference (Bertrand et al. 2004; Brandt and Dlugosch 2021; Donald and Lang 2007; Lam et al. 2022). In estimating the effect of the Ethereum blockchain's transition from PoW to PoS through the Merge, I use Bitcoin as the control group, which adopted PoW as its task assignment mechanism throughout the analysis period in 2022. Also, Bitcoin and Ethereum, the largest and second largest cryptocurrencies used and traded in the world, respectively, are comparable in terms of transaction activities central to task assignment, as well as their market size and social influence (Bitcoinwiki 2019). As Figure 1 shows, the prices of the two cryptocurrencies are strongly correlated, reflecting the comparability of the two blockchains' overall operation and performance (Pagnotta 2022), making Bitcoin an ideal control group for the DiD analysis.

To obtain task validation and transaction data for Ethereum and Bitcoin, I collected information on blockchain and worker behavior from Google Data Analytics, which provides comprehensive data on the validation and trading activities of the two cryptocurrencies. To triangulate and ensure the accuracy of the data, I also leveraged Dune Analytics, Etherscan.io, btc.com, and blockchain.com as supplementary data sources. The cryptocurrency blockchain is a fast-paced environment, with changes triggered by the switch of consensus protocols occurring within a few days or hours. Therefore, to capture the effects of changes in consensus task assignment, I compared task participation and decentralization within 30 days (1 month) before and after the Merge. My initial sample contains information on 20,620 workers. After narrowing it down to the specific analysis window, my final sample contains information on 3676 workers with task validation records from August 15 to October 15, 2022, reflecting a time window of 1 month before and after the Merge (the observation period). The unit of analysis for H1 and H2, regarding participation and decentralization, is at the worker level, with additional analysis at the task level. For H3A and H3B, the unit of analysis is at the worker level, given the focus on worker characteristics as the boundary condition.

4.2 | Outcome Variables

Participation intensity: To measure worker participation on blockchain, I used the number of transaction blocks assigned to a worker as an indicator of their task intensity. Specifically, the variable *Participation intensity* is calculated as the number of validation tasks of a worker, identified by the reward addresses, within a day. This variable is log-transformed to account for the potential skewness.

In robustness checks, I also used alternative measures based on the detailed block information to examine the task-level dynamics of participation. I focused on the scope of participation, which is central to the theoretical conceptualization, by examining the extent to which tasks are assigned to an extended scope of worker participants. Specifically, as the activities of new workers are crucial indicators of the scope of participation at the task level, I examined whether a given task is assigned to a newly entered worker. *New validator* is a binary variable set to one if block validation rewards go to the address (i.e., an encrypted identifier of a blockchain participant) of a worker that appears for the first time, and 0 otherwise.

Task concentration: H2 focuses on the decentralization of task assignment among workers, that is, the extent to which tasks are evenly distributed among workers. At the worker level, tasks received by a given worker relative to the population average ultimately determine the overall shape of the task distribution on a blockchain and, in turn, the degree of task-level decentralization. Hence, I used a variable that captures the absolute difference between the percentage of blocks validated by a focal worker, among all workers in a day, and the population average percentage to denote worker level *Task concentration* calculated as follows:

$$d_{i,t} = \text{abs}\left(p_{i,t} - \overline{p}_t\right)$$

where $d_{i,t}$ denotes the task concentration of focal worker i relative to the population average, in absolute percentage, on day t; $p_{i,t}$ is the percentage of tasks received by focal worker i on day t; and $\overline{p}_t$ is the population average percentage of tasks received by all workers on a blockchain on day t. The measure takes the absolute value, as the theory focuses on the degree of even distribution across workers. If a given worker does not participate in validation tasks on a focal day, this value is missing. It should be noted that *Task concentration* is calculated reversely to decentralization. The smaller the value, the more evenly the tasks are distributed among workers, meaning a higher level of decentralization. The robustness checks also examined the concentration of tasks at the task level, using a similar measure. In addition, I checked the sensitivity of the results to using alternative measures of task concentration without absolute value, and the results remain robust. Additional analyses were also performed to account for the potential effects of large-scale participants with multiple reward addresses due to liquid staking under PoS.

4.3 | Explanatory Variables

Ethereum: I used the dummy variable *Ethereum* to denote the Ethereum blockchain subject to the treatment (i.e., the Merge), in which the task assignment mechanism switched from PoW to PoS. This variable takes a value of one for workers on Ethereum and zero for those on Bitcoin.

PoS: I used the dummy variable *PoS* to denote the time after the Merge, which takes a value of one after September 15, 2022, and otherwise zero. The interaction *Ethereum* × *PoS* indicates the treatment effect of the Merge, that is, the treatment effect of PoS as the task assignment mechanism compared with PoW.

Worker tenure: I used the entry time to measure worker tenure on the blockchain. Specifically, tenure is calculated as the number of tasks taken since a worker's first validation task on the blockchain. Given that tenure is a time-variant characteristic of workers, the analysis for H3A and H3B is at the worker level. This variable is log-transformed to account for potential skewness. All models controlled for worker-fixed effects and time-fixed effects by day.

4.4 | Estimation

As this study leverages the shock of the Merge using a DiD design, the analysis is based on the following model specification to examine the treatment effect of the Merge (switch of task assignment mechanism from PoW to PoS) on worker task participation and decentralization:

$$Y_{i,t} = \beta_i + \beta_e Ethereum_i + \beta_{e,s} Ethereum_i \ x \ PoS_{i,t} + \vartheta T_t + e_{it} \quad (1)$$

where $Y_{i,t}$ represents the outcome variable of *Participation intensity* or *Concentration* for a given task i on day t, β_i is the intercept, β_e indicates the base average of the treatment group for worker i on *Ethereum*, and $\beta_{e,s}$ indicates the treatment effect of $Ethereum_i \times PoS_{i,t}$ for worker i on day t. ϑ is the effect of T_t, a vector of time-fixed effects, through daily dummies.

When worker-fixed effects, δW, are added to the equation, the model can be rewritten as follows:

$$Y_{i,t} = \beta_i + \beta_{e,s} Ethereum_i \times PoS_{i,t} + \vartheta T_t + \delta W + e_{it} \quad (2)$$

To test the moderating effects of worker tenure, I followed the standard practice in recent implementations of DiD designs with moderators through three-way interactions (e.g., Chen et al. 2022; Wu and Zhu 2022) and included all two-way interactions as controls:

$$\begin{aligned} Y_{i,t} = \beta_0 &+ \beta_{e,s,t} Ethereum_i \times PoS_{i,t} \times Tenure_{i,t} \\ &+ \beta_{e,s} Ethereum_i \times PoS_{i,t} + \beta_{e,t} Ethereum_i \times Tenure_{i,t} \quad (3) \\ &+ \beta_{s,t} PoS_{i,t} \times Tenure_{i,t} + \beta_n \ Tenure_{i,t} + \vartheta \mathrm{T}_t + \delta \mathrm{W} + e_{it} \end{aligned}$$

The models were estimated using ordinary least squares (OLS) regressions for a direct interpretation of the magnitudes of the effects.

5 | Results

5.1 | Descriptive Statistics

Before turning to the regression analysis, I first examined the descriptive data on Ethereum to determine whether the data showed patterns consistent with the hypothesized effects of the Merge. Figure 2A illustrates the value of *Participation intensity* per worker, calculated as the log-transformed number of workers involved in transaction validation during the observation period. The figure shows a distinct jump in the number of workers participating in block generation for Ethereum immediately after the

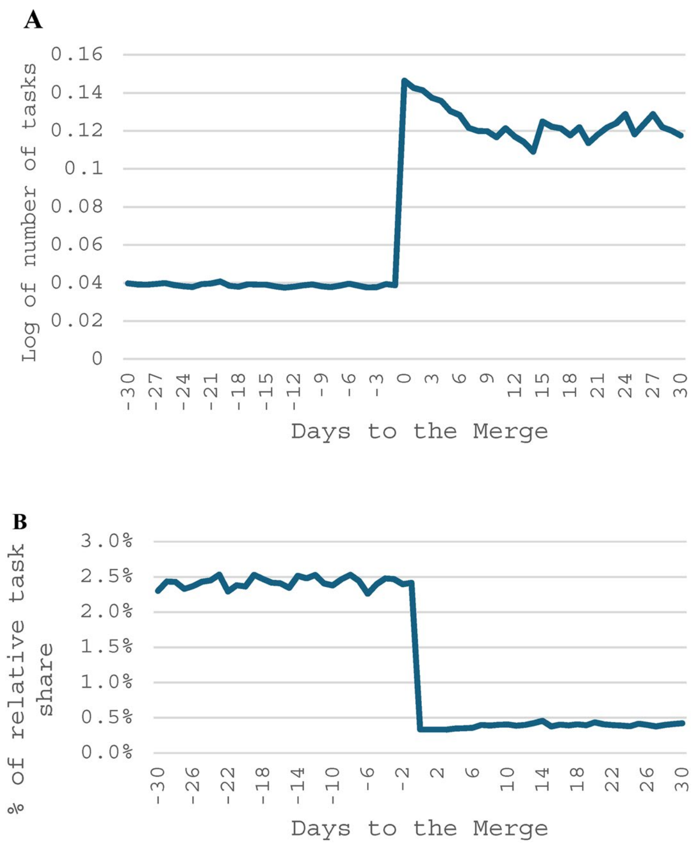

FIGURE 2 | (A) Average *Participation intensity*: Tasks per worker on Ethereum. This figure illustrates participation trends using raw data of Ethereum validation, calculated as the total number of tasks per worker on Ethereum (log-transformed). The x-axis denotes the time to the shock of the Merge, in days. (B) *Task concentration*: Share of task load per worker relative to the population average on Ethereum. This figure illustrates decentralization trends using raw data of Ethereum validation, calculated as the percentage of tasks per worker relative to the population mean. The x-axis denotes the time to the shock of the Merge, in days.

Merge. Similarly, Figure 2B illustrates the centralization tendencies on Ethereum, measured as the relative *Task concentration* of workers during the observation period. The figure shows that for Ethereum, the task concentration drops significantly immediately after the Merge, suggesting that the switch from PoW to PoS has an immediate effect on task decentralization. Together, Figure 2A,B reveal that Ethereum's switch to PoS triggers nonnegligible changes in worker *Participation intensity* and *Task concentration*. The descriptive data show patterns consistent with the hypothesized effects, establishing a basis for examining the causal effects of the Merge and the PoS consensus under the DiD analysis (Fan et al. 2022; Miller et al. 2021; Patel et al. 2021).

5.2 | Main Results and Robustness Tests: H1 and H2

5.2.1 | Assessment of the Parallel Trend Assumption

With the preliminary evidence based on the raw Ethereum data reported above, I next assessed whether the patterns in the data align with the parallel trend assumption, as required by the DiD analysis, in the pre-treatment period with regard to the treated and non-treated observations, that is, whether Ethereum and Bitcoin exhibited similar patterns of movement before Ethereum's transition from PoW to PoS. To do so, I adopted the procedure used in recent research in economics and estimated the dynamic treatment effect by interacting *Ethereum* × *Day* for the periods before and after the Merge (Babina et al. 2023; Miller et al. 2021). Figure 3A,C report the estimates of the dynamic treatment effects between Ethereum and Bitcoin on *Participation intensity* before and after the treatment of the Merge by days, at the worker level (main analysis) and the task level (supplementary analysis) respectively. Each point in the figures denotes the coefficient of the interaction term *Ethereum* × *Day* (pre- or post-treatment), with the vertical line denoting the 95% confidence interval of each coefficient. The figures show that the coefficients denoting the differences between PoW and PoS in *Participation intensity* include zero in all of the pre-Merge days at both the worker and the task level, confirming the pre-treatment parallel trend assumption. Furthermore, they show that *Participation intensity* is significantly higher for the days after the Merge on Ethereum, indicating that the switch from PoW to PoS increased workers' task participation. These patterns are consistent with H1 and support the parallel trend assumption.

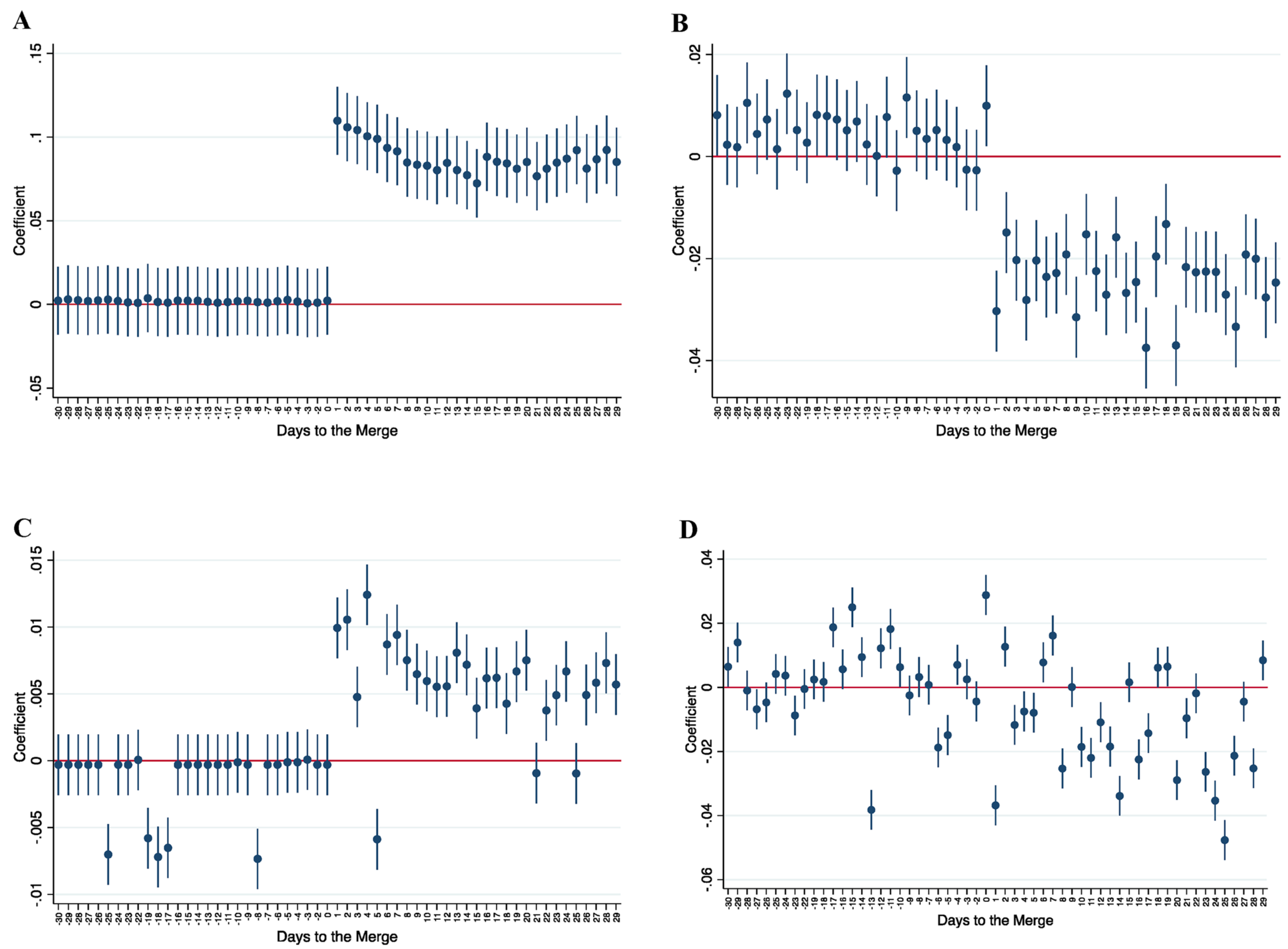

FIGURE 3 | This figure plots the coefficients of the dynamic treatment effects of the Merge with 95% confidence intervals for the days before and after the Merge. (A) Uses *Participation intensity* at the worker level as the DV, and (B) uses *Task concentration* at the worker level as the DV. (C) Uses *Participation intensity* at the task level as the DV, and (D) uses *Task concentration* at the task level as the DV. Each point represents the coefficient of the interaction term.

Similarly, Figure 3B,D plot the dynamic treatment effects through the coefficients of *Ethereum × Day*, with *Task centralization* as the dependent variable at the worker level (main analysis) and the task level (supplementary analysis) in Figure 3B,D, respectively. The figures reveal that Ethereum and Bitcoin do not exhibit consistently significant differences in centralization movement before the treatment of the Merge, confirming the parallel trend assumption. Also, *Task centralization* decreases significantly for Ethereum relative to Bitcoin after the Merge at both the task and worker levels, as suggested by negative and significant coefficients on the average differences in the estimated effects, showing patterns consistent with H2. Together, the coefficients of the dynamic treatment effect reported in Figure 3A–D provide support for the pretreatment parallel trend assumption, with the coefficients of the treatment effects falling into the range of 0 before the Merge for both *Participation intensity* and *Task concentration* at the worker and task levels. The results also suggest that compared with PoW, PoS enables greater worker participation and task decentralization after the Merge, providing preliminary support for H1 and H2.

5.2.2 | DiD Regression Analyses

Table 1 reports the summary statistics of the regression analyses.[6] Table 2 reports the OLS results for the influence of the switch from PoW to PoS task assignment. Models 1 and 3 estimate *Participation intensity* and *Task centralization*, respectively, and are based on Equation (1), without worker-fixed effects; Models 2 and 4 are based on Equation (2), which includes worker-fixed effects. The positive and significant coefficient of *Ethereum × PoS* ($\beta = 0.085$; $p < 0.001$) for Model 2 in Table 2 reveals that *Participation intensity* increases by 8.5% in PoS after the Merge, supporting H1. Furthermore, the negative and significant coefficient of *Ethereum × PoS* ($\beta = -0.029$; $p < 0.001$) for Model 4 of Table 2 reveals that *Task concentration* decreases by 2.9% in PoS after the Merge, indicating a 2.9% reduction in the difference between a focal worker's task share and the average task share of the population and supporting H2. In summary, the DiD analyses reveal that after the Merge, both worker task participation and decentralization on Ethereum increased significantly, supporting H1 and H2, respectively.

5.2.3 | Falsification Analyses of the Treatment Effect

To ensure the accuracy of the main results regarding H1 and H2, I also performed several robustness checks. First, I conducted falsification analyses to ensure that the observed effect of the Merge is not merely a time trend. To do so, I performed a time placebo test by randomly selecting a time before the actual switch as the placebo treatment time, when a treatment effect should not be

TABLE 1 | Descriptive statistics: Worker level.

	Variable	Obs	Mean	Std. dev.	Min	Max	1	2	3	4
1	*Ethereum*	872,178	0.29	0.45	0.00	1.00	1.00			
2	*PoS*	872,178	0.51	0.50	0.00	1.00	0.26	1.00		
3	*Participation intensity*	872,178	0.03	0.26	0.00	7.62	−0.12	−0.46	1.00	
4	*Task concentration*	15,370	0.01	0.03	0.00	0.28	−0.50	−0.33	0.52	1.00
5	*Worker tenure*	15,370	3.15	2.93	0.69	15.00	−0.33	−0.69	0.80	0.50

TABLE 2 | Difference-in-differences estimates of PoS on participation and concentration at the worker level.

	(1)	(2)	(3)	(4)
DV	***Participation intensity***		**Workers' *task concentration***	
Ethereum	0.036***		−0.037***	
	(0.001)		(0.002)	
Ethereum × PoS	0.085***	0.085***	−0.028***	−0.029***
	(0.002)	(0.001)	(0.003)	(0.006)
Constant	0.003***	0.013***	0.068***	0.034***
	(0.000)	(0.000)	(0.002)	(0.005)
Observations	872,178	872,178	15,370	13,492
R-squared	0.028	0.449	0.305	0.936
Time FE	Yes	Yes	Yes	Yes
Worker FE	No	Yes	No	Yes

Note: This table reports the OLS regression results using the DiD design at the worker level. *Ethereum × PoS* denotes the treatment of the Merge, which changed the consensus mechanism from PoW to PoS. *Participation intensity* is measured as the log number of validation tasks per worker, and *Task concentration* is measured by the distance in the share of validation tasks by a worker relative to the population average of the day. Robust standard errors are in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

observed. Table 3 reports the placebo test results, with Models 1, 3, and 5 capturing the effect on validators' *Participation intensity* and Models 2, 4, and 6 capturing the effect on *Task centralization*, based on placebo treatment times of 120 days (Models 1 and 2), 180 days (Models 3 and 4), and 240 days (Models 5 and 6) prior to the actual treatment of the Merge. The non-significant coefficients of *Ethereum × Time Placebo* across the models suggest that the results are not likely to be merely driven by the endogenous time trend, verifying that the observed treatment effects in the main analysis are not artifacts of the natural growth of the two blockchains over time.

5.2.4 | Alternative Time Windows Around Treatment

As discussed above, given the fast-paced nature of cryptocurrency transaction validation (with such tasks being created and assigned every 10–20 min), the effect of the switch from PoW to PoS may manifest within a short time. Indeed, the parallel trend analyses on the dynamic coefficients in Figure 3A–D suggest an immediate effect after the Merge. To check whether the strength of the treatment effect is affected by the time window choice, I tested the robustness of the results by using shorter time windows before and after the Merge. Specifically, I adopted alternative time windows of 7 days and 72 h pre- and post-treatment reported in Models 1–2 and Models 3–4 of Table 4, respectively. The results remain fully robust to these alternative time windows, further supporting H1 and H2.

A related question is the extent to which the stimulating effects of the switch to PoS on task participation and decentralization can be sustained over time after the Merge. While Figures 2A,B and 3A–D all reveal that the effects were sustained over the 30-day window, I rerun the model using a window of 90 days (3 months) before and after the Merge. The results, presented in Models 5 and 6 of Table 4, show consistent evidence demonstrating a stronger effect of *Ethereum × PoS* on both *Participation intensity* ($\beta = 0.003$, $p < 0.001$) and *Task centralization* ($\beta = -0.029$, $p < 0.001$).

Finally, robustness tests were performed to address the possibility that Ethereum and Bitcoin were in different stages of development—as Bitcoin was created more than 6 years before Ethereum, it could lead to different dynamics regarding their workers and tasks. Specifically, blockchains with a longer history may naturally exhibit tendencies toward higher concentration and lower participation due to a higher exit rate, which may create confounding effects. To mitigate this concern, I used an alternative specification of treatment time for Bitcoin, in which the post-treatment dummy variable is set to one for dates after February 20, 2016, which reflects an adjustment to the timeline equivalent to the Merge on Ethereum based on the difference between the two blockchains' creation times. The results reported in Models 7 and 8 of Table 4 remain consistent with the main analyses, helping rule out the potential confounding effects due to the different development stages of the treated and control blockchains.

5.2.5 | Alternative Specification of Outcome Variables

To ensure that the results are not an artifact of the calculation of the outcome variables (*Participation intensity* and *Task concentration*), I performed a sensitivity analysis using alternative measures of the outcome variables. For *Participation intensity*, the main analysis relies on the number of blocks validated by a worker per day as the basis of the measure. In supplemental analysis, I used alternative measures of *Participation intensity*, such as the number of transactions and block size validated by a worker (both measures are log-transformed). The results are reported in Model 1 ($\beta = 0.435$, $p < 0.001$) and Model 2 ($\beta = 0.990$, $p < 0.001$) of Table 5, respectively, which remain consistent with the main results and further support H1.

TABLE 3 | Falsification analyses with randomly chosen treatment times before the actual treatment.

	(1)	(2)	(3)	(4)	(5)	(6)
	120 days prior to the true shock		180 days prior to the true shock		240 days prior to the true shock	
DV	*Participation intensity*	*Task concentration*	*Participation intensity*	*Task concentration*	*Participation intensity*	*Task concentration*
Ethereum × Time Placebo	−0.000	−0.001	0.000	−0.001	0.000	−0.002
	(0.000)	(0.002)	(0.000)	(0.002)	(0.000)	(0.001)
Constant	0.013***	0.031***	0.013***	0.031***	0.014***	0.032***
	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)
Observations	872,178	3775	872,178	3652	872,178	3722
R-squared	0.980	0.942	0.950	0.938	0.975	0.947
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports the DiD OLS regression results at the worker level, based on a hypothetical treatment time, to demonstrate that the results in Table 2 are not random. *Ethereum × Time Placebo* denotes the effect of the hypothetical treatment occurring at a time prior to the actual (true) shock of the Merge. *Participation intensity* is measured as the log of the number of validation tasks per worker, and *Task concentration* is measured as the distance between the share of validation tasks received by a worker and the population average for the day. Robust standard errors are in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

TABLE 4 | Sensitivity analysis: Robustness to time window specifications.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	7 days around the Merge		72h around the Merge		90 days around the Merge		Treatment time since blockchain start	
Time window	***Participation intensity***	**Workers' *task concentration***	***Participation intensity***	**Workers' *task concentration***	***Participation intensity***	**Workers' *task concentration***	***Participation intensity***	**Workers' *task concentration***
Ethereum × *PoS*	0.097***	−0.028***	0.103***	−0.028***	0.003***	−0.029***	0.085***	−0.011***
	(0.003)	(0.007)	(0.004)	(0.010)	(0.001)	(0.006)	(0.001)	(0.002)
Constant	0.013***	0.033***	0.013***	0.034***	0.013***	0.035***	0.013***	0.019***
	(0.001)	(0.006)	(0.001)	(0.008)	(0.000)	(0.004)	(0.000)	(0.001)
Observations	214,470	3111	100,086	1293	2,587,938	17,964	872,178	13,646
R-squared	0.480	0.933	0.526	0.907	0.417	0.929	0.439	0.932
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results using the DiD design at the worker level, based on various time windows around the shock of the Merge. *Ethereum* × *PoS* denotes the treatment of the Merge shock that changed the consensus mechanism from PoW to PoS. *Participation intensity* is measured as the log number of validation tasks per worker, and *Task concentration* is measured by the distance in the share of validation tasks by a worker relative to the population average of the day. Robust standard errors are in parentheses. ***$p<0.01$, **$p<0.05$, *$p<0.1$.

To assess the model's sensitivity to the measure of *Task concentration*, which is calculated in absolute value to reflect the extent of even distribution of tasks, I used alternative measures based on the share of blocks and the share of transaction records validated by a worker in a day, adjusted relative to the baseline level of workload in the 2 months before the shock. The results are reported in Model 3 ($\beta=-0.998$, $p=0.005$) and Model 4 ($\beta=-1.003$, $p=0.011$) of Table 5, respectively, which are consistent with H2.

Another concern is related to the potential effect of business model innovation enabled by PoS, which could affect the true tendencies of participation and decentralization. In particular, the emergence of liquid staking pools and exchanges allows users to deposit small numbers of ETH and collectively establish validator nodes, which may control multiple addresses. This possibility could lead to alternative dynamics, in that even though rewards are assigned to different validators, these validators are related to the same entity in liquid staking pools or exchanges, leading to centralization tendencies despite the observed reduced disparity of tasks of workers as identified by their reward addresses. To address this possibility, I ran an additional test to consider the liquid staking pools of Lido and Coinbase, the two largest liquid staking providers, which account for more than 90% of the market share by that time, through tracing the addresses of smart contracts for distributing block rewards.[7] I then consolidated all validator identifiers related to these major liquid staking providers and recalculated the degree of decentralization and task participation.[8] The results in Models 5 and 6 in Table 5 are based on the alternative measures of the two outcome variables at the liquid staking pool level. The results still show a significant increase in *Participation intensity* ($\beta=0.084$, $p<0.001$) and a decrease in *Task concentration* ($\beta=-0.027$, $p<0.001$) relative to Bitcoin after the Merge, further confirming the robustness of the main results.

5.2.6 | Task-Level Dynamics

Given that the above analyses are based on the validation tasks received by workers, I also investigate task-level dynamics, leveraging task-(block-) level data as the unit of analysis to examine the extent to which a new task of validation goes to a worker currently without a significant share of the workload. Table 6 presents the summary statistics of the variables at the task level. I used whether a new focal block is validated by a new worker to measure the scope of participation, given the cross-sectional structure at the task level in Model 1 of Table 7. *Task concentration* is calculated as the percentage difference between the task share taken by the worker who created the focal block and the blockchain population's average workload on the same day, and the results are reported in Model 2 of Table 7. Table 7 shows that the results of the task-level analysis remain fully consistent with those of the worker-level analysis for *Participation intensity* ($\beta=0.019$, $p<0.001$) and *Task concentration* ($\beta=-0.019$, $p<0.001$), providing additional evidence regarding the worker and task level decentralization.

At the task level, I also checked the robustness of the results to the specification of the dependent variables. For task level participation, I used an alternative measure that reflects whether the task is taken by a worker who has not previously taken a task on the given day (as opposed to being entirely new to the blockchain) in Model 3 of Table 7. The results fully support that tasks

TABLE 5 | Sensitivity analysis: Robustness to alternative measures of participation and concentration.

	(1)	(2)	(3)	(4)	(5)	(6)
	Participation measured as		Concentration measured as		Accounting for liquid staking pools	
DV	Number of transactions	Block size	Absolute % of block tasks	Relative % of transactions	*Participation intensity*	Workers' *task concentration*
Ethereum × PoS	0.435***	0.990***	−0.998***	−1.003**	0.084***	−0.027***
	(0.005)	(0.010)	(0.354)	(0.395)	(0.001)	(0.006)
Constant	0.036***	0.055***	1.117***	1.122***	0.013***	0.032***
	(0.001)	(0.001)	(0.295)	(0.328)	(0.000)	(0.005)
Observations	872,178	872,178	13,492	13,310	872,056	13,485
R-squared	0.374	0.287	0.956	0.952	0.452	0.959
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results using the DiD design at the worker level, based on alternative measures of *Participation intensity* and *Task concentration*. *Ethereum × PoS* denotes the treatment of the Merge, which changed the consensus mechanism from PoW to PoS. Robust standard errors are in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

are more likely to be assigned to workers who have not previously received a task that day under PoS ($\beta = 0.063$, $p < 0.001$), consistent with H1. In terms of *Task concentration*, Model 4 of Table 7 reports the results based on the alternative measure of *Task concentration*, calculated analogously to *Task concentration* at the worker level in Section 5.2.5, using the changes in the share of workload assigned to the worker taking the validation tasks and adjusted based on the worker's task history in the prior 2 months. The significant DiD term *Ethereum × PoS* ($\beta = -0.025$, $p < 0.001$) reveals consistent results that support H2. These analyses further increase confidence that the main results are not an artifact of the time windows and specific measures of the outcome variables.

I also performed task-level robustness checks considering liquid staking. Models 5 and 6 of Table 7 consider the effect of staking pools by consolidating the validator addresses of Lido and Coinbase in the same way as in the worker-level analysis in Models 5 and 6 of Table 5. The results are fully robust to the consideration of liquid staking, supporting H1 and H2 at the task level, even after accounting for the potential concentrating forces of liquid staking pools and exchanges.

To examine the sensitivity of the main results to the specification of time windows at the task level, I performed the same set of analyses, using alternative specifications analogous to the worker-level analysis presented in Table 4. The results, reported in Table 8, are fully consistent with the main models at both the task level and worker level, providing additional support for H1 and H2.

5.3 | Analysis of Heterogeneity Among Validators (H3A and H3B)

A remaining conjecture to test here is whether workers' sensitivity to the changes in task assignment rules may differ based on their characteristics, as argued in H3A and H3B. In other words, the observed effects may stem from certain groups of validators who are more attracted to participation because they are more sensitive to the change from PoW to PoS assignment mechanism. To test H3A and H3B, I estimated the influence on *Participation intensity* and *Task centralization* at the worker level by adding a three-way interaction between worker tenure and the treatment interaction term based on Equation (3). I present the results in Models 1 and 2 of Table 9. In line with H3A, Model 1 shows a negative coefficient of the interaction term *Ethereum × POS × Tenure* ($\beta = -0.171$, $p < 0.001$). Similarly, the interaction term *Ethereum × POS × Tenure*, presented in Model 2, is also significant and negative ($\beta = -0.007$, $p < 0.001$), revealing that the decrease in *Task concentration* is greater for workers with longer tenure. These analyses show that in the case of the Merge, the switch to PoS had a stronger incentivizing effect on newly entered workers, while its dilution effect on task concentration was more pronounced for incumbent workers. In addition, sensitivity analyses using alternative time window specifications of 7 days and 72 h before and after the Merge, as shown in Models 3–4 and Models 5–6 of Table 9, respectively, reveal that such differential effects across workers with shorter and longer tenure sustain in the shorter observation window, suggesting that the change would be immediate upon the switch to PoS. These results further corroborate the empirical analyses at the worker and task levels and support H3A and H3B.

Finally, there is a possibility that workers with shorter or longer tenure experience have different levels of sensitivity to the ETH price (actual cost of acquiring native cryptocurrencies for staking). An increasing native asset price could increase the difficulty for new workers to participate. To explore this conjecture, I examine the relationship between the price of Ether and stake deposit activities on the beacon chain for staking validation tasks. Model 1 of Table 10 presents the results of the effect of ETH price on overall staking, which is not significant. Model 2 presents the results for the association between ETH price and deposits coming from new addresses. The results reveal that higher prices reduce such activities of new validators ($\beta = -0.093$, $p < 0.001$). The results also provide additional evidence supporting the predicted higher sensitivity of new workers to costs as the mechanisms driving H3A and H3B.

TABLE 6 | Descriptive statistics at the task level.

	Variable	Obs.	Mean	Std. dev.	Min.	Max.	1	2	3
1	*Ethereum*	359,692	0.97	0.16	0.00	1.00			
2	*PoS*	359,692	0.54	0.50	0.00	1.00	0.01		
3	*Participation intensity*	359,692	0.00	0.06	0.00	1.00	0.01	0.06	
4	*Task concentration*	359,692	0.09	0.05	0.00	0.18	0.04	−0.12	0.03

TABLE 7 | Difference-in-differences estimates of PoS on task participation and concentration at the task level.

	(1)	(2)	(3)	(4)	(5)	(6)
	Main models		Alternative measures		Accounting for liquid staking	
DV	***Participation intensity* (*new validator*)**	***Task concentration***	***Participation intensity* (new tasks to worker by day)**	***Task concentration* (% of tasks taken by worker of the block)**	***Participation intensity* (*new validator*)**	***Task concentration***
Ethereum	−0.001*	0.020***	−0.090***	−0.016***	0.001***	0.019***
	(0.000)	(0.000)	(0.004)	(0.001)	(0.000)	(0.000)
Ethereum × *PoS*	0.019***	−0.019***	0.063***	−0.025***	0.015***	−0.017***
	(0.001)	(0.001)	(0.006)	(0.001)	(0.001)	(0.001)
Constant	0.001***	0.077***	0.098***	0.149***	0.001***	0.077***
	(0.000)	(0.000)	(0.003)	(0.001)	(0.000)	(0.000)
Observations	359,692	359,692	359,692	359,692	359,692	359,692
R-squared	0.019	0.049	0.023	0.020	0.018	0.049
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results using the DiD design at the task level. *Ethereum* × *PoS* denotes the treatment of the Merge, which changed the consensus mechanism from PoW to PoS. *Participation intensity* (*New validator*) indicates whether a block is assigned to a worker who has not previously taken any validation task. *Task concentration* is measured by the distance in the share of validation tasks by a worker that validated the block relative to the population average of the day. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

6 | Discussion

Viewing blockchain as a distinct organizational form, this study explores how the central governance mechanism, namely, the task assignment incentive design in consensus protocol, affects worker participation and task decentralization. My theoretical foundation builds on the OM technology management literature on IT and blockchain application, and the perspectives of the theory of the firm that emphasizes how asset specificity and transaction cost shape blockchain efficiency as an organizational form (e.g., Hastig and Sodhi 2020; Lumineau et al. 2021; Wang et al. 2022), as well as studies on task assignment in crowdsourcing (e.g., Kadolkar et al. 2024; Liu et al. 2024; Lo et al. 2024; Ta et al. 2021). Focusing on PoS and PoW, the two mainstream consensus in cryptocurrency blockchains with different task assignment specifications, I argue that, in contrast to PoW, which requires participants to commit to task assignment through investment in computing power, PoS incentivizes worker participation and decentralization because the staking requirement of assignment using cryptocurrency with low asset specificity to ensure completion mitigates both the transaction costs of contracting for workers and their tendency toward hyper-competition. This argument is empirically supported by the examination of the Merge upgrade event as a natural experiment on Ethereum in 2022, which switched the consensus of Ethereum from PoW to PoS. I also find that the increase in participation and decrease in task centralization are mainly driven by workers with shorter tenure and an overall redistribution of tasks from longer tenured workers to new entrants. These findings have important implications for consensus design, the broader governance of blockchains, and task assignment in operations that require gig workers or extensive coordination.

6.1 | Managerial Implications

This study provides several managerial implications. First, regarding the development of cryptocurrency systems, the evidence of the incentivizing effect of PoS on worker participation

TABLE 8 | Sensitivity analysis: Robustness to time window specifications for task-level analysis.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Time window	**7 days around the Merge**		**72 h around the Merge**		**90 days around the Merge**		**Treatment time since blockchain start**	
DV	***Participation intensity (new validator)***	***Task concentration***	***Participation intensity (new validator)***	***Task concentration***	***Participation intensity (new validator)***	***Task concentration***	***Participation intensity (new validator)***	***Task concentration***
Ethereum	0.000**	0.017***	0.000	0.021***	−0.001***	0.020***	0.001***	0.225***
	(0.000)	(0.001)	(0.000)	(0.002)	(0.000)	(0.000)	(0.000)	(0.003)
Ethereum × PoS	0.007***	−0.006***	0.009***	−0.013***	0.005***	−0.017***	0.015***	−0.038***
	(0.002)	(0.001)	(0.002)	(0.002)	(0.000)	(0.000)	(0.001)	(0.005)
Constant	0.001*	0.078***	0.001	0.077***	0.001***	0.078***	0.001***	0.525***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.002)
Observations	87,192	87,192	40,579	40,579	1,066,965	1,066,965	359,692	359,692
R-squared	0.004	0.025	0.005	0.040	0.003	0.046	0.018	0.018
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results using the DiD design at the task level. *Ethereum × PoS* denotes the treatment of the Merge, which changed the consensus mechanism from PoW to PoS. *Participation intensity* (*New validator*) measures whether a block is assigned to a worker who has not previously taken any validation task. *Task concentration* is measured by the distance in the share of validation tasks by a worker that validated the block relative to the population average of the day. Robust standard errors are in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

TABLE 9 | Post hoc analysis: The effect of PoS based on worker tenure.

	(1)	(2)	(3)	(4)	(5)	(6)
	Main models		7 days around the Merge		72 h around the Merge	
DV	*Participation intensity*	Workers' *task concentration*	*Participation intensity*	Workers' *task concentration*	*Participation intensity*	Workers' *task concentration*
Ethereum × PoS	0.550**	−0.014***	0.609**	−0.012*	0.284	−0.008*
	(0.248)	(0.004)	(0.301)	(0.006)	(0.395)	(0.004)
Ethereum × PoS × Tenure	−0.171***	−0.007***	−0.152***	−0.007**	−0.131***	−0.008***
	(0.026)	(0.001)	(0.033)	(0.003)	(0.045)	(0.001)
PoS × Tenure	0.001	0.001	−0.014	0.000	−0.010	0.000
	(0.012)	(0.001)	(0.023)	(0.002)	(0.035)	(0.001)
Ethereum × Tenure	0.073	−0.007*	0.053	−0.016***	−0.015	−0.025***
	(0.046)	(0.004)	(0.084)	(0.004)	(0.120)	(0.003)
Tenure	0.173***	0.010***	0.108	0.018***	0.137	0.025***
	(0.040)	(0.003)	(0.080)	(0.003)	(0.116)	(0.002)
Constant	0.611***	0.019***	0.930***	0.023***	1.404***	0.022***
	(0.229)	(0.004)	(0.278)	(0.005)	(0.368)	(0.003)
Observations	13,492	13,492	3111	3111	1293	1293
R-squared	0.941	0.938	0.950	0.941	0.954	0.922
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results using the DiD design considering the heterogeneity of worker tenure. *Ethereum × PoS* denotes the treatment of the Merge, which changed the consensus mechanism from PoW to PoS. *Participation intensity* is measured as the log number of tasks per worker, while *Task concentration* is measured by the absolute differences in the share of tasks by a worker relative to the population mean. *Worker tenure* is measured as the log-transformed number of tasks taken since the worker's first job. Robust standard errors are in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

TABLE 10 | ETH price and staking deposit activities.

	(1)	(2)
DV	Deposits for staking (logged)	New deposit addresses
ETH close price	−0.001	−0.093***
	(0.001)	(0.028)
Constant	10.219***	213.361***
	(0.954)	(45.227)
Observations	59	59
R-squared	0.047	0.108

Note: This table reports the OLS regression results on the effect of Ether price on the staking behavior during the observation periods. Model 1 estimates the influence of the number of Ethers deposited to the beacon chain on Ethereum for staking, and Model 2 estimates the number of workers who made the staking deposit for the first time. Robust standard errors are in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

and task decentralization confirms that PoS can be an effective alternative to PoW in governing blockchain operations without the mining and the huge energy consumption required by the initial design of Bitcoin. In practice, my findings support the promise of emerging blockchains that adopted PoS, such as Ethereum and Solana, which could in turn affect the competition of cryptocurrencies and the dominance of Bitcoin as the early entrant. As PoS-backed cryptocurrency blockchains tend to support more complicated smart contracts, the potential competitive advantage also suggests the viability of more generalized blockchain-based organizations that run on such decentralized public infrastructures (Cong et al. 2023; Zhao et al. 2022). Moreover, the efficiency of staking that uses highly accessible and transferrable task commitment in decentralization also suggests that PoS could be an important step toward solving the impossible triangle of decentralization, security, and efficiency (scalability) in the development of blockchains and cryptocurrencies, as it allows the system to scale while maintaining decentralization without the "rich getting richer effect" in PoW (Cong et al. 2023; Malik et al. 2022; Shang et al. 2023).

Second, this study also has managerial implications for the operation of private permissioned blockchains as a new form of organizing coordination and collaboration (Lumineau et al. 2021; Wang et al. 2022). Although my empirical analysis focuses on permissionless cryptocurrency, the fundamental features and mechanisms driving the hypotheses can also apply to permissioned blockchains (Kumar et al. 2020). Importantly, the theoretical mechanism of this study highlights that the efficiency of PoS in incentivizing workers largely originates from the use of blockchain-native crypto assets as assignment rules and incentives, which can provide insights for firms managing extensive collaboration on tasks within and across organizations in manufacturing, logistics, and supply chains (Babich and Hilary 2020; Sodhi et al. 2022). For example, PoS and its staking setup could be used to organize vendors for procurement (Lumineau et al. 2021) or third-party testing to test for counterfeits in the recycling of electronic products and luxury goods to encourage the participation of complementors while ensuring the quality of their work (Babich and Hilary 2020). More generally, my results imply that a low-asset specificity commitment requirement for task assignment is effective for promoting decentralized collaboration in the absence of a coordinating authority; thus, a PoS-like setup is particularly suitable for managing collaborations among a large number of equally small partners (Hsieh and Vergne 2023; van Rijn et al. 2024).

This study also has generalizable implications for task management with crowd workers and operations with diversified and dispersed tasks because of digitalization, specialization, and globalization (Amaya and Holweg 2024; Kadolkar et al. 2024; Sampson and dos Santos 2023). Although the principles of commitment used in blockchain task assignment have been applied to other crowdsourcing contexts (Castillo et al. 2022; Kaynar and Siddiq 2023; Lo et al. 2024), this study specifically shows that the unique design of the PoS consensus regarding the use of native assets as both commitments and rewards can promote higher worker participation. Such practices of tokens as reward incentives and assignment rules may also apply to centralized contexts such as retail and social media platforms to incentivize the more active exchange of valuable information or services (Benson et al. 2020; Ta et al. 2021; Zeng et al. 2023).

6.2 | Limitations

The findings of this study are subject to a few caveats. The underlying theoretical mechanism is developed at a higher level of abstraction to extend the generalizability of the conceptualization and highlight the nature of blockchain as an organizational form to broaden the applicability to other blockchain practices, but it also suggests that it may simplify the detailed market dynamics of cryptocurrencies in the transaction layer. In addition, as this study focuses on two consensus protocols originally created for cryptocurrency, it does not encompass a comprehensive set of consensus designs, especially those that may have developed outside cryptocurrency. While up to 2025, PoW and PoS are still the two most prevalent consensus protocols, further studies may consider other consensus designs, especially with the increasing practices of DAOs and blockchain applications in other contexts (Hastig and Sodhi 2020; Zhao 2023).

In terms of the nature of the network, a major premise of this study's argument is that a decentralized structure is a fundamental strategic purpose and indicator of the efficiency of blockchain, which calls for caution about generalizability regarding the efficiency conclusion. Beyond the scope of this study, whether decentralization equals efficiency in the presence of centralized authority for coordination remains to be explored by future research. In particular, while other organizational forms like platforms also emphasize tasks decentralization and specialization (Miao et al. 2023; Rietveld and Eggers 2018), the nature of those tasks are essentially complementary activities, and decentralization is intended to reduce complementors' profits and bargaining power and to encourage cross-side network effects (Brandt and Dlugosch 2021; Wen and Zhu 2019). While this study's theoretical foundation draws on research on crowdsourcing that relies on automated task management for gig workers, future studies can explore whether the incentive design of PoS that motivates workers' participation in infrastructure tasks can be applied to complementary tasks in those settings to better understand the scope of the applicability and generalizability of my conclusion.

It should also be noted that the nature of tasks of blockchain validation is relatively homogenous and standardized (Arnosti and Weinberg 2022; Shang et al. 2023). It may also constrain the generalizability of this study because the task itself inherently negates the importance of capability and the costs of production but magnifies the importance of workers' ability to mitigate transaction costs in contracting (Arnosti and Weinberg 2022). While this study shows that less experienced workers benefit more from PoS than seasoned workers, it is worth exploring how competence affects workers' sensitivity to assignment designs of non-standardized and indivisible heterogeneous tasks, which are common in centralized crowdsourcing settings such as TaskRabbit and MTurk (Cullen and Farronato 2021; Kaynar and Siddiq 2023).

Finally, the nature of the blockchain system with anonymous workers could also affect the generalizability of my findings. As anonymity removes the role of workers' reputations, it differs from centralized gig-economy contexts such as ridesharing and content creation platforms that favor workers with high reputations (e.g., Benson et al. 2020; Wu and Zhu 2022; Zeng et al. 2023). As a worker's reputation is also a system-specific commitment but with lower transferability beyond the system, future research may also explore how task assignment based on such nontransferable system commitment incentivizes worker participation to provide more insights into the boundary conditions of my findings.

6.3 | Theoretical Contributions and Opportunities for Future Research

6.3.1 | Research of Technology Management on Blockchain-Based Organizations

This study first deepens the understanding of blockchain by extending the conceptualization of blockchain as a distinct organizational form. Recent literature has increasingly noticed that blockchain constitutes a distinct organizational form (Wang et al. 2022; Zhao et al. 2022) as an alternative to traditional arm's-length and hierarchical structure distinguished by the theory of the firm and transaction cost economics in understanding the coordination of economic activities (e.g., Parkhe 1993; Williamson 1985, 1991). Building on this notion, this study theorizes how consensus serves as a key governance design of blockchain as a distinct organizational form, and how the design shapes the operational efficiency of blockchain system infrastructure by influencing workers' motivation and behavior through the process of task assignment. By exploring the relative efficiency of PoS and PoW in incentivizing task participation and decentralization, I highlight that blockchain shifts the transaction costs of organizing from centralized authorities to distributed crowd workers and that the incentivizing effects of PoS originate from the lower transaction costs for workers in contracting with the system for task assignment, as the asset specificity of the investment required by consensus protocols varies. Moreover, while current research mostly focuses on a single type of consensus through analytical models (Capponi et al. 2023; Roşu and Saleh 2021), this study also contributes to this literature by juxtaposing two consensus protocols, with empirical evidence on the efficiency of PoS based on a natural experiment with research design to derive causal inferences.

The findings of this study provide several directions for future research to explore blockchain organizations. While this study focuses on task participation and decentralization at the worker level, it does not address in detail the heterogeneous nature of workers in terms of their sources of funding to acquire stakes in PoS. Indeed, as considered in this study's supplementary analysis, PoS may promote the growth of liquid staking pools, where participants pool smaller stakes to participate in task assignments, a practice similar to mining pools under PoW that warrants further investigation (Cong, He, and Li 2021). A related direction is the effect of consensus on other actors, such as senders and receivers of cryptocurrencies as blockchain users. The literature has long documented the network externality of blockchain systems, as worker behavior can affect other participants' preferences, and vice versa (Garratt and van Oordt 2023; Shang et al. 2023), and transaction costs exist among the latter during blockchain transactions (Shang et al. 2023). It is worth exploring whether and how the participation and decentralization of transacting parties affect the consensus design and transaction cost of workers at the infrastructure level discussed in this study. Extending this study's highlight of blockchain as an organizational form, future work can further explore factors shaping the choices between blockchain organizations and centralized platforms or traditional alliances in coordinating extensive collaboration (Lumineau et al. 2021; Wang et al. 2022).

This study also advances the understanding of the incentive design of blockchain organizations beyond cryptocurrency. My theorization of consensus from the task assignment perspective explicitly builds on the conceptualization that a blockchain architecture composes two highly connected layers, namely cryptocurrencies at the transaction layer and the operation system at the infrastructure layer (Li et al. 2023). While prior studies largely focus on the dynamics in the transaction layer, this study brings attention to the design and operations of the key infrastructure essential to all blockchain organizations. Following this study, future research could explore the effect of consensus on decentralization in other types of blockchain organizations. In particular, while task participation is directly related to decentralization in cryptocurrencies because rewards incentives representing blockchain rights are automatically distributed with task assignments (Benhaim et al. 2023), other forms of blockchain organizations like DAOs may differ in how decentralization is linked to worker/user participation, with more complex designs of token distribution (Cong et al. 2023; Zhao et al. 2022). In addition, whereas this study focuses on highly liquid major cryptocurrencies, future studies can explore how trading activities at the transaction layer and prices of tokens used as stakes shape how consensus designs affect task participation and decentralization by altering transaction costs on blockchains that vary in levels of activities and liquidity (Pagnotta 2022).

6.3.2 | Research on the Task Assignment of Crowdsourcing Labor

This study also contributes to the understanding of task assignments and the role of incentive design in crowdsourcing. While blockchains present unique features that differ from centralized crowdsourcing, a fundamental connection of my study with this line of literature is the use of commitment (investment requirement) as an incentive design to ensure task completion by fluid workers unconstrained by organizational hierarchy and relational contracts (Li and Zhang 2023; Lo et al. 2024; Pagnotta 2022). In particular, the random assignment under the equal staking of PoS echoes the consideration of fairness highlighted by the task assignment literature on how gig workers' contributions and participation are both extrinsically and intrinsically motivated (e.g., Bai et al. 2022; Kc et al. 2020; Shah 2006; Ta et al. 2021; Tan and Staats 2020). My theorization of PoS on the use of native blockchain currency as both incentives and assignment rules provides additional insights into the interactive dynamics of extrinsic motivation (i.e., the expected pecuniary return net of contracting costs) and intrinsic motivation (i.e., the perception of fairness based on the design of consensus protocols) of crowd workers and reveals that it can jointly shape their contribution behavior in decentralized automated systems. Moreover, as consensus is essentially a case of task assignment that invokes a distinct contracting process between workers and a distributed digital system, this study also advances the understanding of task assignment design for crowdsourcing in more generalized contexts beyond blockchain that leverage other automated assignment rules with highly distributed tasks due to digitalization, specialization, and globalization (Alblas 2022; Spring et al. 2022).

Another contribution of this study to the understanding of task assignments is the explicit consideration of competition among gig workers, which is prevalent in other operational contexts involving crowdsourced tasks and workers. Studies have noted intense competition among workers in gig economies (e.g., when nearby Uber drivers compete for the same ride or when DoorDash or Instacart drivers in a parking lot compete for the same delivery order), which makes task assignment a salient issue for platform governance and competition (Castillo et al. 2022; Sampson and dos Santos 2023; van Rijn et al. 2024). My argument on the demotivating effect of competition due to the contracting cost with highly specified assets brings attention to the role of task assignment design in shaping worker competition. My findings also provide new insights into how crowdsourcing platforms can leverage task assignment mechanisms to strike a balance between attracting workers and reducing hyper-competition through a PoS-like incentive specification (Liu and Tsyvinski 2021; Sockin and Xiong 2023).

Following those directions, future studies could explore the implications of using PoS-like rules in the incentive design of assignments rules and rewards for functional/complementary tasks to govern traditional centralized gig economy platforms. Future studies could also investigate whether the contracting requirement specific to a centralized system can produce the same effect as PoS on blockchains—that is, whether a centralized but tradable cryptocurrency as stakes can be possible for traditional organizations and how it shapes worker incentives and participation behavior in similar or different ways. Given that the literature has suggested that the efficiency of governance and incentive design hinges on the nature of tasks (Nickerson and Zenger 2004), future studies can explore which types of tasks in centralized organizations are more sensitive to PoS-like rules in incentivizing participation. Future research can also investigate how the required number of stakes in centralized platforms influences worker participation.

Moreover, while this study focuses on the pecuniary aspect of incentives, it is worth examining how task assignment rules could be based on nonpecuniary incentives as intrinsic motivation (such as system rank or prior participation intensity), which are more common in centralized settings such as retail, content producing, ridesharing, and research programs (Chung et al. 2023; Gallus 2017; Radaelli et al. 2024; Ta et al. 2021). Extending this study's theorization on the transformation from extrinsic to intrinsic motivation of validators through fairness perception of staking under PoS, future studies may explore the possibility of directly combining pecuniary and nonpecuniary incentives as extrinsic and intrinsic motivation in task assignment in centralized contexts for managing gig workers (e.g., Gallus 2017; Lo et al. 2024; Ta et al. 2021), and how such strategy affects cross-side network effects on the demand side critical to the competitive advantage of centralized crowdsourcing (Liu et al. 2024).

6.3.3 | Research on IT/Algorithm Management

This study also advances the understanding of algorithms and IT discussed in the broader OM literature on technology management (Holmström et al. 2019; Kadolkar et al. 2024). While an extended body of literature has explored how algorithms shape worker behavior in operations, especially the extent to which workers rely on the decision recommendations by algorithms (Castillo et al. 2022; Liu, Tang, et al. 2023; Miao et al. 2023), this study's examination of consensus protocols as a fully automated governance mechanism for blockchain highlights another underexplored implication of algorithms and IT management. The findings of this study indicate that IT/algorithms could play a more substantial role in the efficiency and competitiveness of organizational forms by affecting the accumulation of fundamental human capital that sustains the key operations of the digital infrastructure. Moreover, the moderating effect of experience in shaping worker behaviors of participation and task decentralization identified in this study also extends the growing literature on the behavioral implications of IT-based operations for workers with heterogeneous backgrounds in fully automated settings (Ai et al. 2023; Alblas 2022; Dong and Ibrahim 2020).

While this study focuses on task assignment, future studies could explore the influences of fully automated algorithms in other key operational processes and incentive designs. For example, while validation tasks on blockchains are independent of each other, future studies can investigate management and task assignments using automated programs for interdependent tasks on decentralized blockchains or in other complex IT systems. Moreover, with the development of AI, the decentralized blockchain coordination based on computer programs explored in this study provides opportunities for further studies to investigate how the development of automation technologies, such as deep learning and large language models, may further impact the evolution of consensus and other blockchain-based smart contract design, aiding a better understanding of how technologies can facilitate the diffusion of blockchain as a distinct organizational form in the digital age (Babich and Hilary 2020; Spring et al. 2022; Wang et al. 2022).

Endnotes

[1] https://www.theblock.co/post/292579/bitcoin-surpasses-one-billion-transactions-processed-eight-hundred-weeks-after-launch.

[2] While the literature has suggested that the potential cross-side network effects typical of traditional platforms may also exist in blockchain (e.g., Pagnotta 2022; Sockin and Xiong 2023), this paper focuses on the direct effect of task assignment on workers and relies on the distinct characteristics of blockchain to develop the key hypotheses. The potential cross-side network effects of task assignment on users and transactions provide important clues for future research to explore, which are discussed as future research directions in Section 6.

[3] At the time of the Merge, both Bitcoin and Ethereum largely relied on either program-specific ASIC mining machines (meaning that the chips could only be used to mine a specific PoW blockchain) or GPUs, which are more constrained in supply and more difficult to liquidate for exit (Capponi et al. 2023; Li et al. 2023).

[4] An additional consideration here is that PoS can also facilitate business model innovation such as liquid staking, which allows users to deposit smaller amounts of cryptocurrencies than the staking requirement into liquid staking pools or through centralized exchange platforms, which then provide the crypto funds collectively to entities willing to take on validation tasks on blockchain. The business model innovation of liquid staking should further expand the scope of participants, as it allows users with dispersed funds to indirectly participate in validation, while also enabling entities lacking cryptocurrencies to take part in validation leveraging funds from the liquid staking pool provided by users. It should further lower the participation barrier and expand the base of workers, thereby increasing decentralization. In this study, I focus on the direct effect, without accounting for the indirect effects of users and business model innovation, which therefore warrant further investigation in future studies.

[5] It should also be noted that the number of stakes on mainstream blockchains is relatively low and maintained the same for all participants (e.g., 32 Ethers). However, the actual entry barrier may not necessarily be the number of stakes but, rather, their value. Furthermore, the capped requirement of equal stakes creates the possibility that a natural entity may serve as several validators, particularly those affiliated with liquid staking pools like Lido or centralized exchanges like Coinbase. The blockchain workers referred to in the theory development include the consideration of entities that holds multiple validator qualifications. The empirical analysis also accounts for the possibility that multiple validator addresses could belong to the same entity through robustness checks.

[6] *Worker tenure* has a relatively high correlation with *Participation intensity*. Results are fully robust to using orthogonalized values as an alternative specification, which addresses the concern of potential biases arising from highly correlated variables.

[7] It should be noted that liquid staking differs from traditional entities in terms of control of rewards and individual workers. Liquid staking pools like Lido operate in the form of DAOs with independent validator nodes, whose staking deposits are capped to maintain the market share of each node at a level below 1%. For exchanges like Coinbase, they

basically broker the staking activities between users with dispersed funds and workers lacking sufficient deposits. In this sense, such liquid staking activities would not interfere with decentralization on the worker side but would expand the scope of participation by attracting a higher level of funding sources in a way that would account for the observed greater participation and associated decentralization in the main analysis. In addition, pools exist in the context of mining under PoW; as shown in Models 5 and 6 of Table 5, consolidating workers by liquid staking pools presents relatively conservative estimates that overestimate the concentration under PoS.

[8] https://www.coinbase.com/institutional/research-insights/research/market-intelligence/growth-of-the-liquid-staking-market.

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